Joel SOFFO

Issy-les-Moulineaux, France | (+33) 695302542 | joelpascalsoffo@yahoo.fr |
https://www.linkedin.com/in/joelsoffo | https://github.com/JoelSOFFO | https://joelsoffo.github.io/

Brief description

Machine Learning Scientist with experience in designing AI pipelines for physics-based simulation problems. Currently completing a Ph.D. in AI / Applied Mathematics, with defense scheduled on October, 19 2026 — available immediately for full-time opportunities.

Experience

Machine Learning Scientist (Ph.D. CIFRE) July 2023 – July 2026
Airbus *Issy-les-Moulineaux, France*

- Developed and deployed machine learning models for complex physical simulation problems.
- Developed new dimension-reduction methods that make simulations significantly faster while maintaining the required level of accuracy.
- Mentored an intern (ENSTA Paris) on the application of machine learning to physical simulation, from defining the research topic to presenting the results.
- Invited/contributed talks at national and international conferences.

Machine Learning Scientist Oct. 2022 – June 2023
Airbus *Issy-les-Moulineaux, France*

- Applied research in Scientific Machine Learning (SciML).
- Implemented and evaluated Physics-Informed Neural Operators and Neural Networks (PINNs) for industrial simulation tasks.

Machine Learning Research Intern Apr. 2022 – Sep. 2022
Airbus *Issy-les-Moulineaux, France*

- Evaluated state-of-the-art methods in Scientific Machine Learning (Physics-Informed Neural Networks, Neural Operators).
- Built experimental AI pipelines for physics-based simulations.
- Presented results to multidisciplinary teams.

Data Scientist Intern Apr. 2021 – Sep. 2021
Labelians *Nemours, France*

- Analyzed pricing data across client segments to identify disparities and define a data-driven convergence strategy.
- Designed a statistical pricing model to progressively align disparate client price points toward a unified target over a multi-year horizon.
- Collaborated with multidisciplinary teams to deploy a user-friendly application enabling marketing teams to simulate and track pricing convergence.

Education

Ph.D. in AI / Applied Mathematics 2023 – 2026
Ecole Centrale de Nantes *Nantes, France*

- **Research topics**: Machine Learning for physics problems; Deep Learning; Dimensionality reduction methods; Physics-Informed Neural Networks (PINNs); Neural Operators

Master's degree in Data Science 2020 – 2022
Institut Polytechnique de Paris | Université Paris-Saclay *Palaiseau, France*

- Awarded a two-year academic scholarship by the Fondation Mathématique Jacques Hadamard (FMJH).
- **Courses**: Advanced Deep Learning, Generative models, Advanced Learning for text and graphs, Computer Vision, Statistical Learning Theory.

Publications

- A. Bensalah, A. Nouy, J. Soffo, "Nonlinear manifold approximation using compositional polynomial networks", in *SIAM Journal on Scientific Computing, 2026.*
- M. Dus, G. Dusson, V. Ehrlacher, C. Guillot, J. Soffo, "Comparison between tensor methods and neural networks in electronic structure calculations" in *ESAIM, proceedings and surveys, 2025.*

Certifications

Deploy ML Models to Production | *Kodekloud – Coursera* Sep. 2026

- https://coursera.org/share/48a5b3ff786763e9b86c20e6261ae21f

Introduction to Containers with Docker, Kubernetes and OpenShift | *IBM – Coursera* Sep. 2026

- https://coursera.org/share/83965f9e77bf59838f1d5866ab571b24

Selected Projects

Generative Modeling for Financial Risk | *Python, PyTorch* 2022

- Competed in a generative modeling challenge organized by BNP Paribas and École Polytechnique (X).
- Designed and trained GAN and VAE architectures to simulate financial stress-testing scenarios.
- Awarded Third Prize among 60+ participating teams.
- Presented the solution to BNP Paribas stakeholders at their Paris headquarters.

Ransomware Detection via Bitcoin Transactions | *Python, Scikit-learn, Git* 2022

- Built feature extraction pipeline from blockchain data.
- Trained classification models to detect ransomware attack families.

Languages

French: Native
English: Fluent

Technical Skills

Languages: Python, R, MATLAB, SQL
Frameworks: PyTorch, Scikit-learn, JAX, Tensorflow
Developer Tools: Git, Docker, Kubernetes, VS Code, Visual Studio, PyCharm, IntelliJ